Sheet1

Random numbers and statistical sampling

Statistics

During the course we will often perform the statistical analysis of data, generated in numerical simulations. It is useful to implement *library* functions performing basic statistical analysis, which you can reuse during the whole semester.

Task: Assuming that your sampled data are stored in a C++ **vector<double>**:

1. Write a function to compute the sample average of the measured data,
2. Write a function to compute the sample standard deviation of the measured data.

Test your program by filling a data vector with random data and compute the average and standard deviation of these data.

Expectation values and standard error

The Monte Carlo simulation methods need random numbers. As the simple random number generator rand() in C is not good enough for such applications, we use more sophisticated random generators, either from the C++11 library (see `http://www.cplusplus.com/reference/random/`) or from GSL (see GNU Scientific Library `https://www.gnu.org/software/gsl/doc/html/index.html`). Examples of both can be found on the exercise website.

Task: Consider a population x distributed according to a Gaussian distribution $p(x)$ with mean $\mu = 0$ and standard deviation $\sigma = 1$. For an observable $f(x)$ the expectation value and variance of f on the population x, distributed according to the probability $p(x)$, are defined as

$$\text{expectation value:} \quad \langle f \rangle_p = \int_{-\infty}^{\infty} dx\, p(x) f(x)$$

$$\text{variance:} \quad \sigma_f^2 = \left\langle \left(f - \langle f \rangle_p \right)^2 \right\rangle_p = \langle f^2 \rangle_p - \langle f \rangle_p^2 .$$

For $f(x) = \cos(x)$ perform the following tasks to investigate how $\langle f \rangle_p$ can be estimated by the sample average $\bar{f}$ of samples of size N:

1. Choose 101 values of N between $N = 10$ and $N = 10^6$, which are **equidistant on logarithmic scale** and perform the following task for *each value* of N:

   a) To *estimate* the expectation value $\langle \cos \rangle_p$, generate a sample with N random numbers distributed according to $p(x)$, i.e. by drawing N random values x_i,

$i = 1 \dots N$, using a random number generator producing numbers distributed according to the distribution $p(x)$. The average $\bar{f}$ of the measurements $f(x_i)$ is an estimate for $\langle f \rangle_p$.

b) For each value of N, repeat this sampling M times (take $M = 100$) on *independent* samples, each of size N. For *each sample*, measure the average $\bar{f}$ for $f(x) = \cos(x)$ as described in (a) and store these M measured values of $\bar{f}$ in a data file, as you will have to plot them later.

2. The above task will yield $101 \times M$ values of $\bar{f}$ (M values for each N). Use gnuplot (`http://www.gnuplot.info`) or another graphics package (e.g. Matplotlib in Python) to:

   a) Plot all the averages $\bar{f}$ as a function of N. There should be M data points for each N. Use **logarithmic scales** on the plots where appropriate.

   b) Use some math program of your choice (WolframAlpha, Mathematica, Matlab, Maple) to compute the exact expectation value $\langle \cos \rangle_p$ and show this value in the plot too.

   c) What do you conclude about the numerical estimates as N is increased?

   d) What would happen if M is increased?

   Hint: Use the example gnuplot script on the exercise page. Export the plots to pdf-files to include in your report.

3. a) Quantify the spread of the *sample averages* $\bar{f}$ for each N by computing the sample standard deviation $s_{\bar{f}}(N)$ of the M computed *averages* $\bar{f}$ for *each value of* N. Make a plot of this standard deviation $s_{\bar{f}}$ as a function of N on a log-log scale.

   b) Fit the function aN^b to the data for $s_{\bar{f}}$ using, for example, gnuplot's fit function (to get a good fit, it is important to use similar relative errors on all data points when fitting). Plot the fit function in the plot too.

   c) Interpret the values of a and b. What drives the coefficient a and what does b tell you about the behavior of the simulation as N is increased?

4. **Write a Latex report** describing your results, including the requested plots.